CALCULUS · UNIT 9: AP CALCULUS REVIEW

Water Park Engineering

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students engineer a water park in AP free-response style: a table of pump rates (Riemann sums, MVT, FTC), raft motion on a lazy river, area and volume of a wave pool, and a chlorine differential equation.

The project doubles as a cumulative review with AP-style justification practice.

Standards Addressed

- AP CHA-4 Accumulation from tables; Riemann sums
- AP CHA-3 Particle motion
- AP CHA-5 Area and volume
- AP FUN-7 Differential equations; AP justification practices

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a released AP FRQ. Part 1 pool table.
Day 2	Part 2 raft motion.
Day 3	Part 3 wave pool, chlorine, and portfolio.

Materials

- Student handout and Pump Rate Table
- Graphing calculators
- AP scoring guidelines (for justification language)

How the Three Versions Differ

Level	What changes
CP2	Left Riemann sum, motion with a linear velocity, basic area and volume, and one AP-style sentence.
CP1	Trapezoidal sum, MVT on a table, trig velocity, area between curves, and an exponential DE.
Honors	Accumulation function, candidates test, two-raft motion with IVT, semicircle cross sections, equilibrium DE, and writing an FRQ.

Watch For (Common Misconceptions)

- Missing units or using the wrong units (gal vs. gal/min).
- Using velocity instead of speed for "speed increasing".
- Forgetting to justify with a theorem or sign change.

Extension Ideas

- Time yourself on a released AP FRQ.
- Trade your FRQ with another group and grade it.
- Research real water park engineering.

Answer Key – CP2

Part 1: Filling the Pool

#	Task	Answer
1	Use a left Riemann sum with the four intervals to estimate the water pumped in from t = 0 to t = 60.	10(120) + 15(150) + 15(175) + 20(160) = 1,200 + 2,250 + 2,625 + 3,200 = 9,275 gal
2	Find the total water drained from t = 0 to 60.	100 × 60 = 6,000 gal
3	Estimate the amount of water in the pool at t = 60.	2,000 + 9,275 − 6,000 = 5,275 gal
4	Estimate R′(30) using the table and give units.	$\frac{160-175}{40-25}$ = −1 gal/min per minute

Part 2: Lazy River Raft

#	Task	Answer
5	When is the raft at rest? When is it moving backward?	t = 3; backward for 0 ≤ t < 3
6	Find the acceleration.	a(t) = 2 ft/min²
7	Find the displacement from t = 0 to t = 8.	$\int_0^8 (2t-6)\,dt = 64 - 48 = 16$ ft
8	Find the total distance traveled.	$\int_0^3 \lvert 2t-6\rvert\,dt + \int_3^8 (2t-6)\,dt = 9 + 25 = 34$ ft

Part 3: Wave Pool

#	Task	Answer
9	Find the surface area of the wave pool.	$\int$ from −3 to 3 of $(9 - x^2)\,dx = 36$ m²
10	The depth is 2 m everywhere. Find the volume of water.	72 m³
11	Write one AP-style sentence that explains what $\int_0^{60} R(t)\,dt$ means.	It is the total number of gallons pumped into the pool from t = 0 to t = 60 minutes.

Reflection

#	Prompt	Look for
R1	Why are units so important on AP free-response questions?	They show what the number means (rate vs. amount).
R2	How is displacement different from total distance?	Displacement counts backward motion as negative; distance counts all motion as positive.

Answer Key – CP1

Part 1: Filling the Pool

#	Task	Answer
1	Estimate $\int_0^{60}$ R(t) dt with a trapezoidal sum.	$\frac{10}{2}(270) + \frac{15}{2}(325) + \frac{15}{2}(335) + \frac{20}{2}(290)$ = 1,350 + 2,437.5 + 2,512.5 + 2,900 = 9,200 gal
2	Find the amount of water at t = 60 if the pool starts with 2,000 gal.	Drained $\int_0^{60}$ (100 + 0.5t) dt = 6,900; W(60) ≈ 2,000 + 9,200 − 6,900 = 4,300 gal
3	Is the amount of water increasing or decreasing at t = 40? Justify.	R(40) − D(40) = 160 − 120 = 40 > 0 → increasing
4	Must there be a time in (10, 40) when R′(t) = $\frac{1}{3}$? Justify. (Assume R is differentiable.)	MVT: $\frac{160-150}{30} = \frac{1}{3}$; yes

Part 2: Lazy River Raft

#	Task	Answer
5	When is the raft moving right? Left?	Right 0 < t < 6; left 6 < t < 12
6	Find the acceleration at t = 4 and whether the speed is increasing or decreasing.	a(4) = $\frac{\pi}{2}\cos(\frac{2\pi}{3})$ ≈ −0.785; v(4) ≈ 2.6 > 0, a < 0 → speed decreasing
7	Find the position at t = 6 and the maximum distance from the dock.	x(6) = $\frac{36}{\pi}$ ≈ 11.46 ft (maximum)
8	Find the total distance traveled from 0 to 12.	$\frac{72}{\pi}$ ≈ 22.9 ft

Part 3: Wave Pool & Chlorine

#	Task	Answer
9	Find the area of the region.	Intersections x = −3, 2: ∫ from −3 to 2 of (6 − x − x²) dx = $\frac{125}{6}$ ≈ 20.8 m²
10	Cross sections perpendicular to the x-axis are rectangles with depth 2 m. Find the volume.	2 × $\frac{125}{6} = \frac{125}{3}$ ≈ 41.7 m³
11	Chlorine decays by $\frac{dC}{dt}$ = −0.2C (ppm per hour) with C(0) = 3. Solve and find when C = 1 ppm.	C = 3e^(−0.2t); t = 5 ln 3 ≈ 5.5 hours

Reflection

#	Prompt	Look for
R1	How do you justify that an amount is increasing on the AP exam?	Show that its rate of change (in − out) is positive at that time.
R2	When can you use the MVT on a table?	When the function is differentiable, the average rate between two points equals R′ somewhere between them.

Answer Key – Honors

Part 1: Pool Accumulation

#	Task	Answer
1	Write W(t) as an expression involving an integral.	$W(t) = 2{,}000 + \int_0^t (R(x) - D(x))\,dx$
2	Estimate W(60) using a right Riemann sum for R.	Right: 10(150) + 15(175) + 15(160) + 20(130) = 9,125; W ≈ 2,000 + 9,125 − 6,900 = 4,225 gal
3	Find the time on [0, 60] when W is greatest, given R(t) − D(t) > 0 on (0, 52) and < 0 on (52, 60). Justify.	t = 52 by the candidates test / sign change of W′ from + to −
4	Find the average rate of draining over 0 ≤ t ≤ 60.	$\frac{6{,}900}{60}$ = 115 gal/min

Part 2: Raft Motion

#	Task	Answer
5	Find all times in (0, 12) when the first raft's speed is increasing.	v and a same sign: $a = \frac{\pi}{2}\cos(\frac{\pi t}{6})$; 0 < t < 3 (v > 0, a > 0) and 6 < t < 9 (v < 0, a < 0)
6	Find the distance between the rafts at t = 6.	$x(6) = \frac{36}{\pi} \approx 11.46$; y(6) = 7.2 − 6 = 1.2 → ≈ 10.26 ft
7	Is there a time when the rafts have the same velocity? Justify with the IVT.	y′ = 0.4t − 1. At t = 0: 0 vs −1; at t = 6: v = 0 vs 1.4. h = v − y′ changes from + (1) to − (−1.4), so yes.
8	Find the average velocity of the first raft from t = 0 to t = 9.	$\frac{x(9) - x(0)}{9} = \frac{18}{\pi} \div 9 = \frac{2}{\pi} \approx 0.64$ ft/min

Part 3: Design Your Own FRQ

#	Task	Answer
9	Find the volume if cross sections perpendicular to the x-axis are semicircles.	$\frac{\pi}{8}\int$ from −3 to 2 of $(6 - x - x^2)^2\,dx = \frac{\pi}{8} \times \frac{625}{6} \approx 40.9$ m³
10	Find the equilibrium chlorine level and solve with C(0) = 3. Is the level safe (1 to 3 ppm) in the long run?	C = 1.5 + 1.5e^(−0.2t) → 1.5 ppm; yes
11	Write and solve your own four-part AP-style question about the park, with a scoring guide.	Answers vary.

Reflection

#	Prompt	Look for
R1	What makes a justification complete on the AP exam?	It names the theorem or test, states the conditions, and refers to the given functions with values.

#	Prompt	Look for
R2	How do the big ideas of calculus (rates and accumulation) connect in this project?	Rates (derivatives) describe change, and integrals of rates give accumulated amounts.